HW5 Class Demo Notes

Guided hints for random-effects ANOVA, MLE, REML, and shrinkage

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Summary

1. SSE estimates only within-group noise.
2. SSA estimates between-group variation plus averaged noise.
3. MLE and REML differ mainly in how they account for estimating the fixed effect.
4. Random-effect predictions are shrunken versions of group fixed-effect estimates.

Data preview: the motivating question

Use the `pulp` data as the concrete example.

```
data(pulp, package = "faraway")
head(pulp)
```

```
  bright operator
1   59.8        a
2   60.0        a
3   60.8        a
4   60.8        a
5   59.8        a
6   59.8        b
```

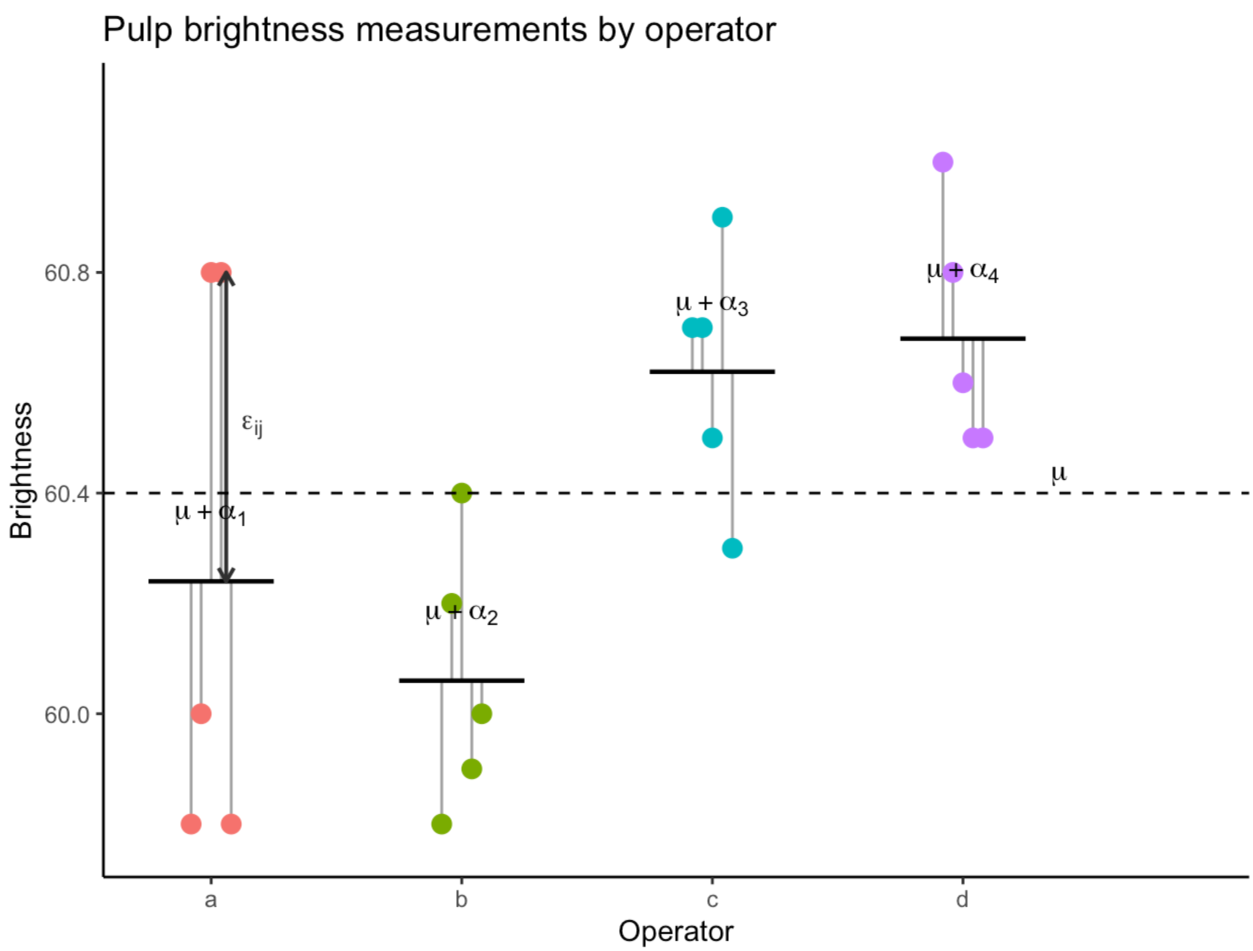
```
pulp |>
  group_by(operator) |>
  summarize(
    n = n(),
    mean_bright = mean(bright),
    .groups = "drop"
  )
```

```
# A tibble: 4 × 3
  operator     n mean_bright
  <fct>    <int>       <dbl>
1 a            5        60.2
2 b            5        60.1
3 c            5        60.6
4 d            5        60.7
```

```r
pulp_plot <- pulp |>
  group_by(operator) |>
  mutate(
    operator_id = as.integer(operator),
    x_jitter = operator_id + seq(-0.08, 0.08, length.out = n()),
    mean_bright = mean(bright)
  ) |>
  ungroup()

pulp_means <- pulp_plot |>
  distinct(operator, operator_id, mean_bright) |>
  mutate(
    mean_label = paste0("mu + alpha[", row_number(), "]"),
    label_y = mean_bright + 0.12
  )

overall_mean <- mean(pulp$bright)
overall_mean_label_x <- length(levels(pulp$operator)) + 0.35

within_variation_label <- pulp_plot |>
  slice_max(abs(bright - mean_bright), n = 1, with_ties = FALSE) |>
  mutate(
    label_x = x_jitter + 0.12,
    label_y = (bright + mean_bright) / 2
  )

ggplot(pulp_plot, aes(x = x_jitter, y = bright, color = operator)) +
  geom_segment(
    aes(xend = x_jitter, y = mean_bright, yend = bright),
    color = "gray65",
    linewidth = 0.5,
    show.legend = FALSE
  ) +
  geom_segment(
    data = pulp_means,
    aes(
      x = operator_id - 0.25,
      xend = operator_id + 0.25,
      y = mean_bright,
      yend = mean_bright
    ),
    inherit.aes = FALSE,
    color = "black",
    linewidth = 0.8
  ) +
  geom_point(size = 3) +
  geom_segment(
    data = within_variation_label,
    aes(
      x = x_jitter + 0.06,
```

```
      xend = x_jitter + 0.06,
      y = mean_bright,
      yend = bright
    ),
    inherit.aes = FALSE,
    color = "gray20",
    linewidth = 0.7,
    arrow = grid::arrow(
      ends = "both",
      length = grid::unit(0.08, "inches")
    )
  ) +
  geom_text(
    data = within_variation_label,
    aes(x = label_x, y = label_y),
    label = "epsilon[ij]",
    color = "gray20",
    hjust = 0,
    parse = TRUE,
    inherit.aes = FALSE
  ) +
  geom_text(
    data = pulp_means,
    aes(x = operator_id, y = label_y, label = mean_label),
    color = "black",
    parse = TRUE,
    inherit.aes = FALSE,
    show.legend = FALSE
  ) +
  geom_hline(yintercept = overall_mean, linetype = 2) +
  annotate(
    "text",
    x = overall_mean_label_x,
    y = overall_mean,
    label = "mu",
    parse = TRUE,
    hjust = 0,
    vjust = -0.4
  ) +
  scale_x_continuous(
    breaks = seq_along(levels(pulp$operator)),
    labels = levels(pulp$operator),
    expand = expansion(mult = c(0.05, 0.22))
  ) +
  scale_y_continuous(expand = expansion(mult = c(0.08, 0.15))) +
  labs(
    x = "Operator",
    y = "Brightness",
    title = "Pulp brightness measurements by operator"
  ) +
  coord_cartesian(clip = "off") +
```

```
  theme_bw() +
  theme(
    legend.position = "none",
    plot.margin = margin(5.5, 34, 5.5, 5.5),
    axis.line = element_line(colour = "black"),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.border = element_blank(),
    panel.background = element_blank()
  )
```



Model setup

The balanced one-way random-effects model is

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \qquad i = 1, \dots, a, \quad j = 1, \dots, n,$$

where

$$\alpha_i \overset{iid}{\sim} N(0, \sigma_\alpha^2), \qquad \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\epsilon^2).$$

Here:

- μ is the overall mean.
- α_i is the random effect for group/operator i.
- σ_α^2 is the between-group variance.
- σ_ϵ^2 is the within-group variance.

Part 1: ANOVA estimators

Step 1: Start with the two means

Ask students to write the group mean:

$$\bar{y}_{i\cdot} = \frac{1}{n}\sum_{j=1}^{n} y_{ij}.$$

Using the model,

$$\bar{y}_{i\cdot} = \mu + \alpha_i + \bar{\epsilon}_{i\cdot}.$$

The grand mean is

$$\bar{y}_{\cdot\cdot} = \frac{1}{a}\sum_{i=1}^{a} \bar{y}_{i\cdot} = \mu + \bar{\alpha}_{\cdot} + \bar{\epsilon}_{\cdot\cdot}.$$

Step 2: Derive the expectation of SSE

The within-group sum of squares is

$$SSE = \sum_{i=1}^{a}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot})^2.$$

Now subtract the group mean:

$$\begin{aligned} y_{ij} - \bar{y}_{i\cdot} &= (\mu + \alpha_i + \epsilon_{ij}) - (\mu + \alpha_i + \bar{\epsilon}_{i\cdot}) \\ &= \epsilon_{ij} - \bar{\epsilon}_{i\cdot}. \end{aligned}$$

So the group effect disappears. What does SSE measure after μ and α_i cancel? Answer: only within-group noise.

For a fixed group i,

$$\sum_{j=1}^{n}(\epsilon_{ij} - \bar{\epsilon}_{i\cdot})^2$$

is the usual sample sum of squares from n iid observations with variance σ_ϵ^2. Therefore

$$E\left[\sum_{j=1}^{n}(\epsilon_{ij} - \bar{\epsilon}_{i\cdot})^2\right] = (n-1)\sigma_\epsilon^2.$$

There are a groups, so

$$E(SSE) = a(n-1)\sigma_\epsilon^2.$$

Therefore

$$\widehat{\sigma}_\epsilon^2 = \frac{SSE}{a(n-1)} = MS_E.$$

Step 3: Derive the expectation of SSA

The between-group sum of squares is

$$SSA = \sum_{i=1}^{a}\sum_{j=1}^{n}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2.$$

Since the expression inside does not depend on j,

$$SSA = n\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2.$$

Now use

$$\bar{y}_{i\cdot} = \mu + \alpha_i + \bar{\epsilon}_{i\cdot}.$$

Define

$$u_i = \alpha_i + \bar{\epsilon}_{i\cdot}.$$

Then

$$\bar{y}_{i\cdot} = \mu + u_i,$$

and

$$\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = u_i - \bar{u}_{\cdot}.$$

Now compute the variance of u_i:

$$\begin{aligned} Var(u_i) &= Var(\alpha_i + \bar{\epsilon}_{i\cdot}) \\ &= Var(\alpha_i) + Var(\bar{\epsilon}_{i\cdot}) \\ &= \sigma_\alpha^2 + \frac{\sigma_\epsilon^2}{n}. \end{aligned}$$

For iid $u_1, \ldots, u_a$,

$$E\left[\sum_{i=1}^{a}(u_i - \bar{u}_{\cdot})^2\right] = (a-1)Var(u_i).$$

Therefore

$$E(SSA) = n(a-1)\left(\sigma_\alpha^2 + \frac{\sigma_\epsilon^2}{n}\right)$$
$$= (a-1)(n\sigma_\alpha^2 + \sigma_\epsilon^2).$$

So

$$MS_A = \frac{SSA}{a-1}$$

has expectation

$$E(MS_A) = n\sigma_\alpha^2 + \sigma_\epsilon^2.$$

Since

$$E(MS_E) = \sigma_\epsilon^2,$$

we get

$$\widehat{\sigma}_\alpha^2 = \frac{MS_A - MS_E}{n}.$$

Step 4: ANOVA estimator summary

The ANOVA estimators are

$$\widehat{\mu} = \bar{y}_{\cdot\cdot},$$

$$\widehat{\sigma}_\epsilon^2 = MS_E = \frac{SSE}{a(n-1)},$$

and

$$\widehat{\sigma}_\alpha^2 = \frac{MS_A - MS_E}{n} = \frac{SSA/(a-1) - SSE/[a(n-1)]}{n}.$$

Pulp data: compute ANOVA quantities by hand

```
y <- pulp$bright
g <- pulp$operator

a <- nlevels(g)
n <- as.integer(table(g)[1])

group_means <- tapply(y, g, mean)
grand_mean <- mean(y)

SSE <- sum((y - ave(y, g))^2)
SSA <- sum(as.numeric(table(g)) * (group_means - grand_mean)^2)

MS_E <- SSE / (a * (n - 1))
```

```
MS_A <- SSA / (a - 1)

sigma2_e_anova <- MS_E
sigma2_a_anova <- (MS_A - MS_E) / n

c(
  a = a,
  n = n,
  grand_mean = grand_mean,
  SSE = SSE,
  SSA = SSA,
  MS_E = MS_E,
  MS_A = MS_A,
  sigma2_e_anova = sigma2_e_anova,
  sigma2_a_anova = sigma2_a_anova
)
```

```
             a              n     grand_mean            SSE            SSA 
    4.00000000     5.00000000    60.40000000     1.70000000     1.34000000 
          MS_E           MS_A sigma2_e_anova sigma2_a_anova 
    0.10625000     0.44666667     0.10625000     0.06808333 
```

Part 2: MLE hints and derivation

Step 1: Write the marginal distribution for one group

For group i, write the vector

$$\mathbf{y}_i = \begin{pmatrix} y_{i1} \\ \vdots \\ y_{in} \end{pmatrix}.$$

Because

$$\mathbf{y}_i = \mu \mathbf{1}_n + \alpha_i \mathbf{1}_n + \boldsymbol{\epsilon}_i,$$

we have marginally

$$\mathbf{y}_i \sim N(\mu \mathbf{1}_n, V),$$

where

$$V = \sigma_\alpha^2 \mathbf{1}_n \mathbf{1}_n^T + \sigma_\epsilon^2 I_n.$$

For $j \neq k$,

$$Cov(y_{ij}, y_{ik}) = \sigma_\alpha^2.$$

For $j = k$,

$$Var(y_{ij}) = \sigma_\alpha^2 + \sigma_\epsilon^2.$$

Step 2: Matrix facts for the likelihood

The log-likelihood uses V^{-1} and $\det(V)$.

By the Woodbury formula,

$$V^{-1} = \sigma_\epsilon^{-2} I_n - \frac{\sigma_\epsilon^{-2}\sigma_\alpha^2}{\sigma_\epsilon^2 + n\sigma_\alpha^2}\mathbf{1}_n\mathbf{1}_n^T.$$

Also,

$$\det(V) = \sigma_\epsilon^{2n}\left(1 + n\frac{\sigma_\alpha^2}{\sigma_\epsilon^2}\right).$$

Now define

$$\lambda = \frac{\sigma_\alpha^2}{\sigma_\epsilon^2}.$$

This gives

$$\det(V) = \sigma_\epsilon^{2n}(1 + n\lambda).$$

Step 3: Profile over μ

The log-likelihood is maximized in μ at

$$\widehat{\mu} = \bar{y}_{\cdot\cdot}.$$

After setting $\mu = \bar{y}_{\cdot\cdot}$, the likelihood can be written using only SSE and SSA.

Step 4: Profile over σ_ϵ^2

After reparameterizing by λ, the profiled estimate of σ_ϵ^2 satisfies

$$\widehat{\sigma}_\epsilon^2(\lambda) = \frac{SST + n\lambda\, SSE}{na(1 + n\lambda)}.$$

Equivalently,

$$\widehat{\sigma}_\epsilon^2(\lambda) = \frac{SST - \frac{n\lambda}{1+n\lambda} SSA}{na}.$$

The only remaining unknown is λ.

Step 5: Maximize over λ

After profiling out μ and σ_ϵ^2, the likelihood reduces to maximizing

$$-\frac{na}{2}\log(SST + n\lambda\, SSE) + \frac{(n-1)a}{2}\log(1+n\lambda).$$

Taking the derivative with respect to λ and setting it equal to zero gives

$$\widehat{\lambda} = \frac{n-1}{n}\frac{SST}{SSE} - 1.$$

Substituting this back gives

$$\widehat{\sigma}^2_{\epsilon,MLE} = \frac{SSE}{a(n-1)}.$$

So the MLE and ANOVA estimators for σ^2_ϵ agree in this balanced case.

For the between-group variance,

$$\widehat{\sigma}^2_{\alpha,MLE} = \frac{SSA}{an} - \frac{SSE}{an(n-1)}.$$

Compare this with ANOVA:

$$\widehat{\sigma}^2_{\alpha,ANOVA} = \frac{SSA}{(a-1)n} - \frac{SSE}{a(n-1)n}.$$

MLE is usually smaller because it uses a instead of $a-1$ for the between-group component. It does not make the same degrees-of-freedom correction.

Pulp data: MLE calculation

```r
SST <- sum((y - grand_mean)^2)

sigma2_e_mle <- SSE / (a * (n - 1))
sigma2_a_mle <- SSA / (a * n) - SSE / (a * n * (n - 1))

c(
  SST = SST,
  sigma2_e_mle = sigma2_e_mle,
  sigma2_a_mle = sigma2_a_mle,
  sigma2_e_anova = sigma2_e_anova,
  sigma2_a_anova = sigma2_a_anova
)
```

```
           SST   sigma2_e_mle   sigma2_a_mle sigma2_e_anova sigma2_a_anova
    3.04000000     0.10625000     0.04575000     0.10625000     0.06808333
```

Part 3: REML hints

Step 1: The conceptual issue

MLE estimates variance components after also estimating μ. REML adjusts for that lost fixed-effect degree of freedom.

Here the fixed-effect design matrix is just

$$X = \mathbf{1}_{na}.$$

So REML removes the direction spanned by $\mathbf{1}_{na}$.

Step 2: The projection-like matrix

Let

$$\Omega = \sigma_\alpha^2 ZZ^T + \sigma_\epsilon^2 I.$$

The REML likelihood involves

$$P = \Omega^{-1} - \Omega^{-1}X(X^T\Omega^{-1}X)^{-1}X^T\Omega^{-1}.$$

This matrix keeps variation orthogonal to the fixed effect.

Step 3: Balanced one-way conclusion

In the balanced one-way random-effects model, REML gives the same variance component estimates as ANOVA:

$$\widehat{\sigma}^2_{\epsilon,REML} = MS_E,$$

and

$$\widehat{\sigma}^2_{\alpha,REML} = \frac{MS_A - MS_E}{n}.$$

REML agrees with ANOVA here because both methods account for the degree of freedom used to estimate the overall mean.

Part 4: Posterior mean of random effects

Now consider the conditional model

$$\mathbf{y} \mid \boldsymbol{\gamma} \sim N(X\beta + Z\boldsymbol{\gamma}, \sigma^2 I),$$

and the prior

$$\boldsymbol{\gamma} \sim N(0, \Sigma).$$

Step 1: Multiply likelihood by prior

By Bayes' theorem,

$$f(\boldsymbol{\gamma} \mid \mathbf{y}) \propto f(\mathbf{y} \mid \boldsymbol{\gamma}) f(\boldsymbol{\gamma}).$$

Ignoring constants, the exponent is

$$-\frac{1}{2\sigma^2}\|\mathbf{y}-X\beta-Z\boldsymbol{\gamma}\|^2-\frac{1}{2}\boldsymbol{\gamma}^T\Sigma^{-1}\boldsymbol{\gamma}.$$

Step 2: Collect quadratic and linear terms

Let

$$\mathbf{r}=\mathbf{y}-X\beta.$$

Then

$$\|\mathbf{r}-Z\boldsymbol{\gamma}\|^2=\mathbf{r}^T\mathbf{r}-2\boldsymbol{\gamma}^TZ^T\mathbf{r}+\boldsymbol{\gamma}^TZ^TZ\boldsymbol{\gamma}.$$

Keeping only terms involving $\boldsymbol{\gamma}$, the negative exponent is proportional to

$$\frac{1}{2}\boldsymbol{\gamma}^T\left(\sigma^{-2}Z^TZ+\Sigma^{-1}\right)\boldsymbol{\gamma}-\boldsymbol{\gamma}^T\sigma^{-2}Z^T(\mathbf{y}-X\beta).$$

Therefore the posterior precision matrix is

$$\sigma^{-2}Z^TZ+\Sigma^{-1}.$$

So the posterior covariance is

$$\left(\sigma^{-2}Z^TZ+\Sigma^{-1}\right)^{-1}.$$

The posterior mean is

$$E(\boldsymbol{\gamma}\mid\mathbf{y})=\left(\sigma^{-2}Z^TZ+\Sigma^{-1}\right)^{-1}\sigma^{-2}Z^T(\mathbf{y}-X\beta).$$

Using the matrix inversion identity, this is equivalent to

$$E(\boldsymbol{\gamma}\mid\mathbf{y})=\Sigma Z^T\left(Z\Sigma Z^T+\sigma^2I\right)^{-1}(\mathbf{y}-X\beta).$$

Step 3: Specialize to balanced one-way ANOVA

For the one-way random-effects model,

$$\Sigma=\sigma_\alpha^2I_a,\qquad\sigma^2=\sigma_\epsilon^2.$$

Also,

$$Z^TZ=nI_a.$$

The posterior mean is

$$\begin{aligned}E(\boldsymbol{\alpha}\mid y)&=\left(\sigma_\epsilon^{-2}nI_a+\sigma_\alpha^{-2}I_a\right)^{-1}\sigma_\epsilon^{-2}Z^T(y-\mu\mathbf{1})\\&=\frac{\sigma_\alpha^2}{\sigma_\epsilon^2+n\sigma_\alpha^2}Z^T(y-\mu\mathbf{1}).\end{aligned}$$

For group i,

$$Z_i^T(y - \mu\mathbf{1}) = \sum_{j=1}^{n}(y_{ij} - \mu) = n(\bar{y}_{i\cdot} - \mu).$$

So

$$E(\alpha_i \mid y) = \frac{n\sigma_\alpha^2}{\sigma_\epsilon^2 + n\sigma_\alpha^2}(\bar{y}_{i\cdot} - \mu).$$

Equivalently,

$$E(\alpha_i \mid y) = \frac{1}{1 + \sigma_\epsilon^2/(n\sigma_\alpha^2)}\hat{\alpha}_i,$$

where

$$\hat{\alpha}_i = \bar{y}_{i\cdot} - \mu.$$

The multiplier

$$\frac{1}{1 + \sigma_\epsilon^2/(n\sigma_\alpha^2)}$$

is between 0 and 1.

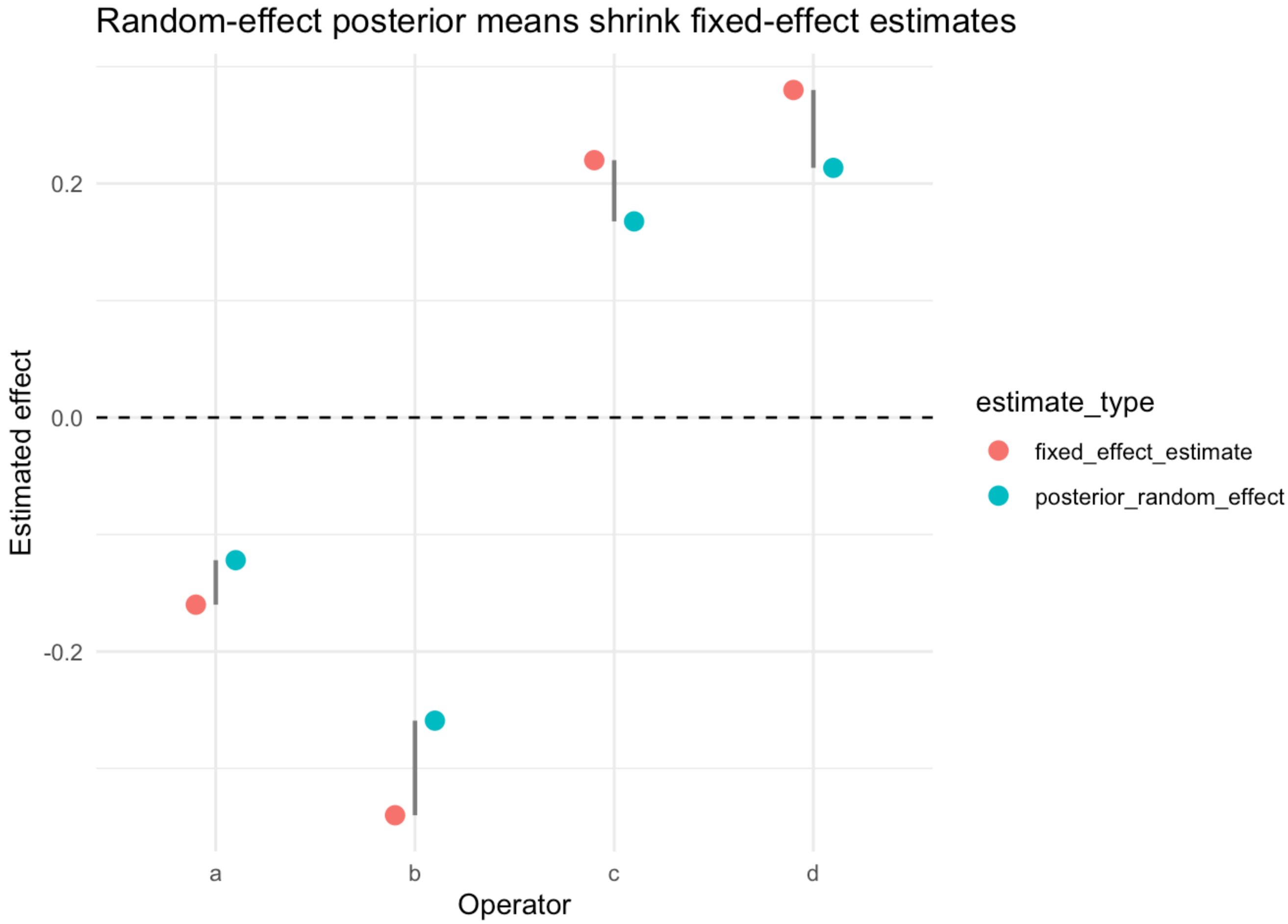
An extreme group mean may reflect a true group effect, but it may also reflect noise. The model compromises by pulling it toward the overall mean.

Pulp data: visualize shrinkage

```
shrinkage_factor <- n * sigma2_a_anova / (n * sigma2_a_anova + sigma2_e_anova)

effects <- tibble(
  operator = names(group_means),
  fixed_effect_estimate = as.numeric(group_means - grand_mean),
  posterior_random_effect = shrinkage_factor * fixed_effect_estimate
)

effects
```

```
# A tibble: 4 × 3
  operator fixed_effect_estimate posterior_random_effect
  <chr>                    <dbl>                   <dbl>
1 a                       -0.160                  -0.122
2 b                       -0.340                  -0.259
3 c                        0.220                   0.168
4 d                        0.280                   0.213
```

```r
effects_long <- effects |>
  pivot_longer(
    cols = c(fixed_effect_estimate, posterior_random_effect),
    names_to = "estimate_type",
    values_to = "effect"
  )

ggplot(effects_long, aes(x = operator, y = effect, color = estimate_type)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_point(size = 3, position = position_dodge(width = 0.4)) +
  geom_segment(
    data = effects,
    aes(
      x = operator,
      xend = operator,
      y = fixed_effect_estimate,
      yend = posterior_random_effect
    ),
    inherit.aes = FALSE,
    color = "gray50",
    linewidth = 0.8
  ) +
  labs(
    x = "Operator",
    y = "Estimated effect",
    title = "Random-effect posterior means shrink fixed-effect estimates"
  ) +
  theme_minimal()
```



Final class summary

Use this as the closing slide or board summary:

1. Within groups:

$$SSE \Rightarrow \sigma_\epsilon^2.$$

2. Between group means:

$$SSA \Rightarrow n\sigma_\alpha^2 + \sigma_\epsilon^2.$$

3. ANOVA and REML correct for estimating the overall mean:

$$\widehat{\sigma}_\alpha^2 = \frac{MS_A - MS_E}{n}.$$

4. MLE is similar, but usually estimates the between-group variance smaller.

5. Posterior random effects are shrunken fixed-effect estimates:

$$E(\alpha_i \mid y) = c\hat{\alpha}_i, \qquad 0 < c < 1.$$

One-sentence takeaway: SSE measures noise, SSA measures group variation plus noise, and random-effect predictions shrink noisy group differences toward the grand mean.